Non-invasive Wearable Blood Glucose Monitor

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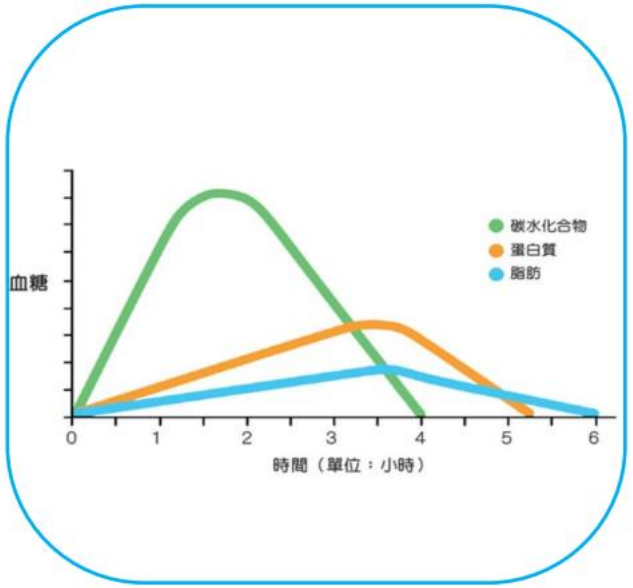
Observe Glucose Curve After Eating (or Drinking Sugary Beverages)

After warm-up, you may eat or drink sugary beverages to observe glucose changes.

Glucose response has a delay (approximately 15–30 minutes) depending on the individual and the type of food.

Key Point: Long-term monitoring reveals how diet affects glucose levels, helping you understand metabolic status and whether glucose fluctuations occur.



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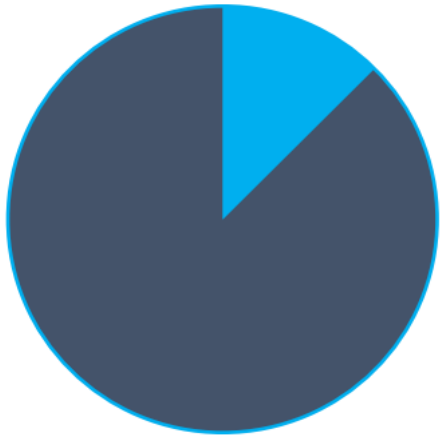
Accuracy Range (MARD ~15%)



- Normal glucose range for healthy individuals: 100–150 mg/dl
- MARD (15%) — Mean Absolute Relative Difference
- Considering MARD ~15%, the display range for a normal person (including potential error) will fall between 85–172.5 mg/dl

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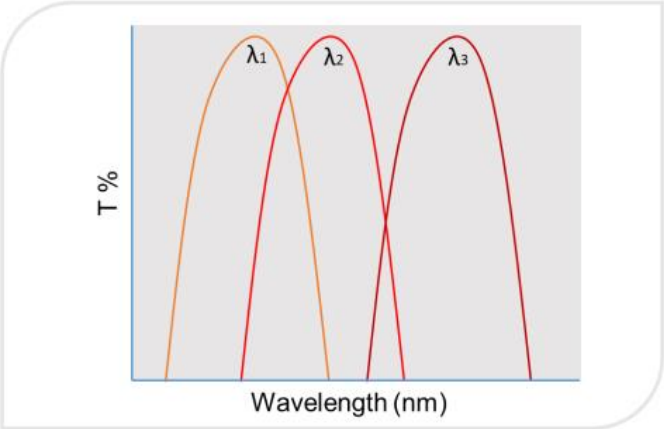
Accuracy Improvement Strategy



Hardware: HUSD & MBPF **Software: Algorithm Enhancement** **System: Noise Suppression**

- Using multi-band-pass filters (MBPF) to suppress unnecessary wavelengths and increase HUSD brightness to improve sensing accuracy.
- Continue algorithm research to improve glucose trend feature extraction.
- Device companies must improve noise suppression for both optical and thermal noise, which is essential for handling environmental variability.

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Core Points

- Individuals with normal glucose values 100–150 mg/dl are the primary test subjects.

- NICGM focuses on continuous long-term monitoring, which provides real-time wellness feedback.
- Single-point measurements only reflect the current glucose level.
- Continuous and long-term data reveals more physiological insights from glucose trends.
- Current objective: allow users to see real-time glucose changes to increase interest in NICGM usage.
- Pediatric growth clinics are potential users—excess sugar intake in children may cause attention issues, hyperactivity, or affect development.
- With an aging population, glucose monitoring becomes increasingly important to reduce risks of chronic diseases and improve quality of life.

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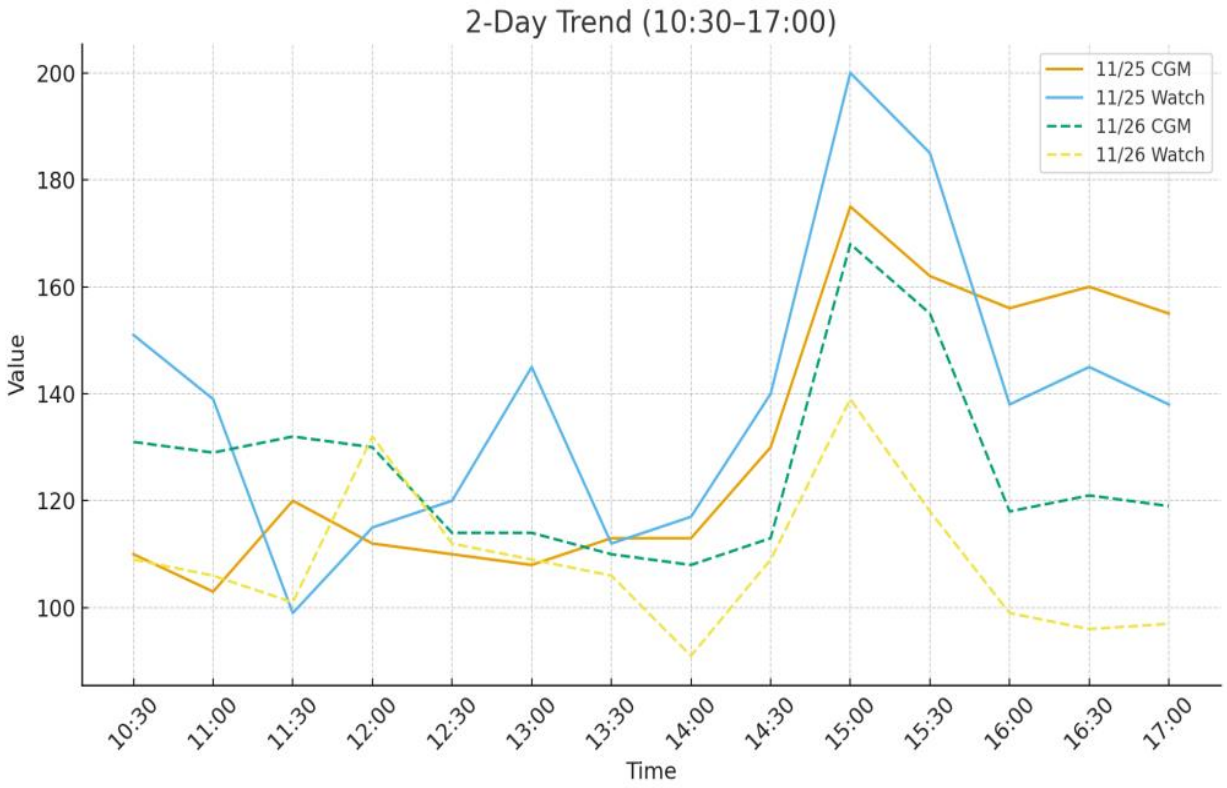
NICGM / CGM Glucose Trend Comparison for a Wearable Client



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